

This enables the rapid prediction of $\Delta\tilde{E}_p(a)$, a fundamental input parameter in solute-strengthening theories, providing critical insights for predicting material strength in multi-component alloys.

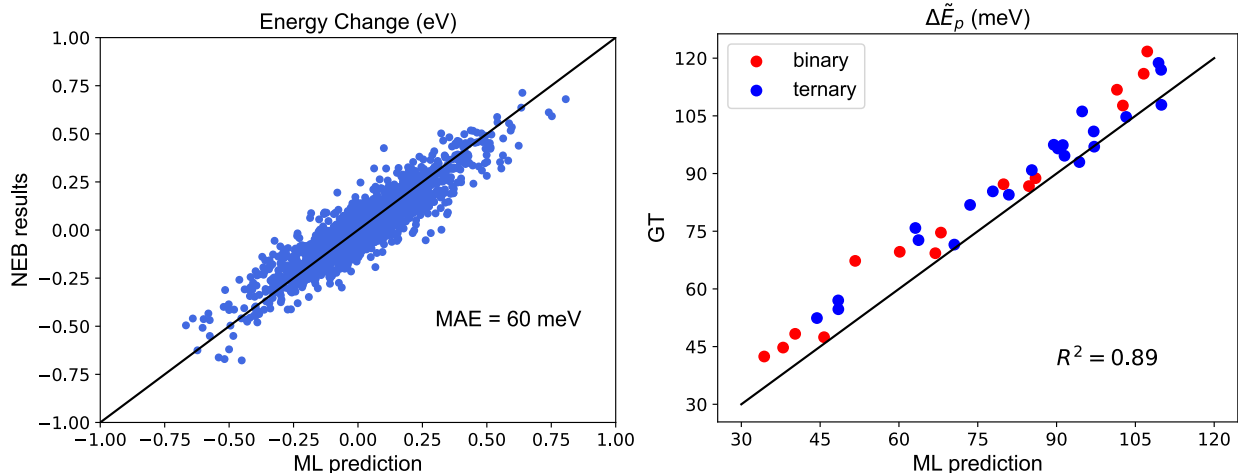


Figure 3: **Evaluation of the GNN model for the potential energy change prediction on the test set.** Test set contains new compositions that never appeared in the training set. (a) Comparison of ML predictions and NEB results for the potential energy change. (b) Comparison of the ML results and NEB results for the solute/screw interaction energy parameter, Eq. 1 of binary and ternary compositions in the test set. The correlation coefficient R^2 between the predictions and the ground truth is 0.89.

2.2 Multi-modal multi-agent model

Figure 4 outlines our multi-agent model for rapid multi-component alloy design and analysis. Our model builds upon AtomAgents [52], a multi-modal multi-agent system designed to extract new physical insights from atomistic simulations. In this version, AtomAgents has been enhanced with our new graph neural network model that predicts fundamental physics data. This advancement offers an efficient way to explore the massive compositional space of multi-component alloys compared to costly atomistic simulations, allowing for solving more challenging problems and rapid design of alloys with enhanced or desired material properties.

In the current multi-agent framework, the core agents are the "User" and the "AI Assistant," with the User posing the queries and the AI Assistant responding by utilizing various tools. These tools are designed to handle a wide range of tasks, from planning and coding to multi-modal data analysis. Additionally, these tools engage a set of autonomous agents that collaborate to respond effectively to the queries provided by the Assistant. A crucial component of this toolset is the physics tools, which empower the model by leveraging both newly developed Graph Neural Network (GNN) models and physics-based theoretical frameworks, such as solute-strengthening theories. These theories are essential for connecting atomistic-level features to macroscopic properties like yield stress. Each agent in our multi-agent system serves a specialized role, defined by a unique profile, and is powered by advanced general-purpose language models from the GPT family, accessed via the OpenAI API. A detailed description of each agent's profile and the implementation details of the AI agents and tools are provided in Materials and Methods.

The tools integrated within our multi-agent system enable the resolution of complex challenges in multi-component alloy design, allowing for automated exploration of the vast compositional design space to identify candidates with superior or targeted performance. These tools are automatically activated by the AI Assistant to perform various tasks throughout the alloy design and analysis process. A general workflow of our model is demonstrated in Figure 5. When the User submits a query, the Assistant calls the planning tool to generate a comprehensive plan, detailing the necessary steps, identifying the required tools, and specifying their input parameters. This planning process is driven by two key agents: the planner and the reviewer. The planner drafts the initial version of the plan, while the reviewer critiques and refines it through continuous iterations. This collaborative process ensures that the final plan is precise and thoroughly detailed, which is crucial as the entire problem-solving process depends on this plan.

Once the plan is finalized, the AI Assistant executes it, typically involving tasks such as predicting material properties using the physics tools, collecting and organizing data, and passing them to the coding tool for